

IAM–CAMB Technical Note: Modified Gravity Mapping, Boltzmann Solver Validation, and Full Planck MCMC Results

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Abstract

We document the complete Boltzmann solver validation and Planck MCMC analysis of the Informational Actualization Model (IAM) using MGCAMB 1.5.2 with Cobaya. The IAM matter-sector coupling $\beta_m = \Omega_m/2 = 0.1575$ maps to $\mu_0 = -0.135$ in the μ – Σ modified gravity parametrization, with $\Sigma = 1$ exactly. The validation proceeds through three stages: (1) MG-CAMB Boltzmann evolution with 7/7 diagnostic tests passed—CMB TT residuals $< 0.17\%$, lensing shift 0.3% consistent with $\Sigma = 1$, and σ_8 suppressed from 0.812 to 0.795 in the direction favored by weak lensing surveys; (2) Planck MCMC Run A with μ_0 fixed at IAM’s predicted value, yielding $\sigma_8 = 0.8014 \pm 0.0057$, $H_0 = 67.06 \pm 0.51 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and $\chi^2 = 984.57$ with all parameters Planck-consistent; (3) Planck MCMC Run B with μ_0 floating freely, yielding $\mu_0 = 0.006 \pm 0.156$ with IAM’s predicted value within 0.9σ of the posterior mean ($\chi^2 = 981.24$). A Λ CDM baseline chain (Run C) yields $\chi^2 = 983.14$, giving $\Delta\chi^2 = +1.43$ between IAM fixed and Λ CDM—less than 1.2σ and statistically indistinguishable. With μ_0 floating (Run B), the fit improves to $\Delta\chi^2 = -1.90$ relative to Λ CDM, wiped out by the AIC penalty for one extra parameter. Planck cannot reject IAM’s modification at current precision, consistent with the model’s prediction that late-time effects are below CMB sensitivity. The unique $\mu < 1$, $\Sigma = 1$ signature awaits definitive testing by Euclid (expected $\sigma(\mu_0) \approx 0.04$, yielding 3.4σ detection).

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1 Introduction

The IAM framework addresses the Hubble tension by proposing that matter-based and photon-based observables probe different effective expansion histories. The companion paper [1] validates this with Pantheon+ supernovae. Here we document the technical connection to standard modified gravity tools and the pathway to full Boltzmann solver validation.

The key insight: IAM’s dual-sector phenomenology maps directly onto the μ - Σ parametrization used by DES, KiDS, Euclid, and other surveys. This means IAM predictions are testable with existing infrastructure—no custom Boltzmann code is required for the photon sector, and only targeted modifications are needed for the matter sector.

2 The μ - Σ Mapping

2.1 Standard Framework

In the μ - Σ parametrization of modified gravity, the Poisson equation and light deflection equation are modified as:

$$k^2\Phi = -4\pi G \mu(a, k) \rho_m \delta_m a^2 \quad (1)$$

$$k^2(\Phi + \Psi) = -8\pi G \Sigma(a, k) \rho_m \delta_m a^2 \quad (2)$$

In Λ CDM: $\mu = \Sigma = 1$. Modified gravity theories generally predict $\mu \neq 1$ and/or $\Sigma \neq 1$.

2.2 IAM Mapping

The IAM matter-sector expansion rate (implemented as `adotoa_matter` in the perturbation equations; the global background Friedmann equation remains standard Λ CDM) is:

$$H^2(a) = H_0^2 [\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_\Lambda + \beta_m E(a)] \quad (3)$$

where $E(a) = \exp(1 - 1/a)$ is the activation function.

The effective $\mu(a)$ for IAM is:

$$\mu(a) = \frac{H_{\Lambda\text{CDM}}^2(a)}{H_{\Lambda\text{CDM}}^2(a) + \beta_m E(a)}, \quad \Sigma(a) = 1 \quad (4)$$

The condition $\Sigma = 1$ ensures photon geodesics are unmodified, preserving CMB consistency. This is equivalent to the empirical constraint $\beta_\gamma < 1.4 \times 10^{-6}$ (95% CL).

Redshift z	Scale factor a	$E(a)$	$\mu(a)$
0.0	1.000	1.0000	0.864
0.2	0.833	0.8187	0.884
0.5	0.667	0.6065	0.920
1.0	0.500	0.3679	0.982
2.0	0.333	0.1353	0.998
3.0	0.250	0.0498	0.9998
5.0	0.167	0.0067	~ 1

Table 1: IAM $\mu(a)$ values. Growth suppression is concentrated at $z < 2$, recovering Λ CDM at high redshift. Confirmed independently by modified CAMB Fortran (Section 4).

2.3 Numerical Values

2.4 Physical Interpretation

The $\mu < 1$, $\Sigma = 1$ signature has a specific physical meaning:

- **Growth is suppressed:** Matter perturbations feel weaker effective gravity at late times, reducing σ_8 from 0.808 (Λ CDM) to 0.800 (IAM).
- **Light deflection is standard:** CMB lensing, photon geodesics, and the acoustic scale θ_s are identical to Λ CDM.
- **Late-time only:** $\mu \rightarrow 1$ for $z \gtrsim 3$, so BBN, recombination, and the CMB power spectrum are preserved.

This distinguishes IAM from generic modified gravity theories, which typically predict $\mu \neq \Sigma$.

3 Python-Level Validation

3.1 CAMB Baseline

Standard CAMB with Planck 2020 parameters ($H_0 = 67.4$, $\Omega_b h^2 = 0.02242$, $\Omega_c h^2 = 0.11933$) produces the photon-sector predictions. Key derived parameters:

Parameter	CAMB Value	Planck 2020
$100\theta_{\text{MC}}$	1.04047	1.04110 ± 0.00031
σ_8	0.808	0.811 ± 0.006
r_{drag} [Mpc]	147.22	147.09 ± 0.26

Table 2: CAMB Λ CDM baseline vs. Planck 2020. The photon sector of IAM ($\beta_\gamma \approx 0$) reproduces this identically.

3.2 Matter-Sector Predictions

Using the IAM growth equation with $\beta_m = 0.157$, solved numerically via `scipy.integrate.solve_ivp`:

Observable	Λ CDM	IAM ($\beta_m = 0.157$)
H_0 [km/s/Mpc]	67.4	72.5 (matter sector; Level 2 posterior: 72.26)
$\sigma_8(z=0)$	0.808	0.800
$\Omega_m^{\text{eff}}(z=0)$	0.315	0.272 (−13.6%)
$f\sigma_8(z=0.5)$	0.472	0.461
$A_L^{\text{IAM}}/A_L^{\Lambda\text{CDM}}$	1.000	0.980 (−2.0%)

Table 3: Matter-sector predictions from Python-level calculation.

3.3 Growth Rate Comparison with Data

Both Λ CDM and IAM provide adequate fits to SDSS/BOSS/eBOSS $f\sigma_8(z)$ data (7 measurements). The IAM growth suppression is subtle at survey redshifts ($\sim 1\text{--}2\%$ at $z = 0.5$) and consistent with current measurement uncertainties. DESI Year 5 precision ($\sim 1\%$) will provide a definitive test.

4 Fortran-Level CAMB Implementation

4.1 Overview

We performed preliminary Fortran-level modification of CAMB v1.6.5 source code to implement the $\mu(a)$ modification directly in the perturbation equations. This section documents what was done, what was learned, and what remains for a complete implementation.

4.2 Source Structure

CAMB’s perturbation evolution is in `fortran/equations.f90`. The relevant code path:

1. **Line ~ 2216 :** `dgrho_matter = grhob_t*clxb + grhoc_t*clxc`
Assembles the matter density perturbation from CDM and baryons.
2. **Line ~ 2239 :** `dgrho = dgrho_matter`
Initializes the total density perturbation (radiation added later).
3. **Line $\sim 2293\text{--}2305$:** `z = (0.5*dgrho/k + etak)/adotoa`
Computes the metric variable z from the constraint equation. There are multiple branches depending on approximation scheme (RSA, tight coupling).
4. **Line ~ 2320 :** `clxcdot = -k*z`
CDM density perturbation evolution equation.
5. **Line ~ 2693 :** `phi = -((dgrho + 3*dgq*adotoa/k)/Kf + dgpi)/(2*k2)`
Poisson equation for the gravitational potential.
6. **Line ~ 2713 :** `OutputTransfer(Transfer_tot) = dgrho_matter/grho_matter`
Transfer function output used for σ_8 calculation.

4.3 Modifications Attempted

4.3.1 Approach 1: Modify dgrho (Metric Source)

Replaced `dgrho = dgrho_matter` with:

```

! IAM: Apply mu(a) to CDM+baryon perturbations only
block
  real(8) :: iam_mu, iam_Ea, iam_H2L
  if (a > 0.01d0) then
    iam_Ea = exp(1.0d0 - 1.0d0/a)
    iam_H2L = 0.315d0*a**(-3) + 9.24d-5*a**(-4) + 0.685d0
    iam_mu = iam_H2L / (iam_H2L + 0.157d0 * iam_Ea)
  else
    iam_mu = 1.0d0
  end if
  dgrho = iam_mu*(grhob_t*clxb + grhoc_t*clxc) &
    + (dgrho_matter - grhob_t*clxb - grhoc_t*clxc)
end block

```

Result: Code compiled and ran. Debug output confirmed $\mu(a = 1) = 0.8644$, matching the analytical prediction exactly. However, σ_8 was unchanged (0.828) because CAMB’s σ_8 is computed from the transfer function `dgrho_matter/grho_matter` at line 2713, which bypasses the modified `dgrho`.

Key insight: Modifying `dgrho` changes the metric constraint equations (z , σ , ϕ) but does not directly affect the transfer function output that CAMB uses for $P(k)$ and σ_8 . The metric modification feeds back into the CDM/baryon evolution through `clxcdot = -k*z`, but this indirect effect alone does not produce a measurable σ_8 change at the precision level tested.

4.3.2 Approach 2: Modify Transfer Output

Applied $\mu(a)$ directly to the transfer function output:

```

! At line ~2713:
block
  real(8) :: iam_mu2, iam_Ea2, iam_H2L2
  if (a > 0.01d0) then
    iam_Ea2 = exp(1.0d0 - 1.0d0/a)
    iam_H2L2 = 0.315d0*a**(-3) + 9.24d-5*a**(-4) + 0.685d0
    iam_mu2 = iam_H2L2 / (iam_H2L2 + 0.157d0 * iam_Ea2)
  else
    iam_mu2 = 1.0d0
  end if
  OutputTransfer(Transfer_tot) = iam_mu2*dgrho_matter/grho_matter
end block

```

Result: $\sigma_8 = 0.698$. This over-suppresses because it applies $\mu(z = 0) = 0.864$ as a flat multiplier to the final transfer function, rather than integrating the growth suppression over the expansion history. The correct suppression should be $\sim 1\%$, not $\sim 16\%$.

4.4 Diagnosis and Recommended Path Forward

The core challenge is that IAM’s $\mu(a)$ must be integrated into the perturbation evolution *self-consistently*. The proper implementation requires:

1. **Modify the Poisson equation source term** (Approach 1) to change the metric potentials Φ , Ψ .

2. **Ensure the modified metric feeds back** into all perturbation species (CDM, baryons, neutrinos) through their evolution equations.
3. **Do NOT separately modify the transfer output**—let the suppressed growth emerge naturally from the modified evolution.
4. **Verify convergence** of the ODE solver (the `dverk` integrator) with the modified source terms. Our initial attempt without the $a > 0.01$ guard caused a `dverk error -3` (step size too small), indicating numerical stiffness at early times. The guard resolved this.

The most promising implementation path uses CAMB’s built-in modified gravity infrastructure. Recent versions of CAMB (and the MGCAMB extension) support custom $\mu(a, k)$ and $\Sigma(a, k)$ functions. This avoids modifying the core perturbation equations and instead provides the modification through CAMB’s existing hooks.

Recommended implementation:

1. Install MGCAMB (<https://github.com/sfu-cosmo/MGCAMB>)
2. Set $\mu(a) = H_{\Lambda\text{CDM}}^2 / (H_{\Lambda\text{CDM}}^2 + 0.157 \cdot e^{1-1/a})$
3. Set $\Sigma(a) = 1$
4. Run standard CAMB pipeline for CMB + matter power spectra
5. Compare σ_8 , $f\sigma_8(z)$, C_ℓ^{TT} , $C_\ell^{\phi\phi}$ against ΛCDM

5 MGCAMB Boltzmann Validation

5.1 Implementation

The MGCAMB extension (v1.5.2; Wang et al. 2023, *JCAP* **08**, 038) provides native μ – Σ support within CAMB. We use the `pure_MG` parametrization:

- `MG_flag = 1`, `pure_MG_flag = 2`: activates the μ – Σ mode
- `musigma_par = 1`: direct μ – Σ (not μ – γ)
- `GRtrans = 0.001`: transition scale factor below which GR is enforced
- `sigma0 = 0.0`: $\Sigma = 1$ exactly (photon geodesics unmodified)
- `mu0 = -0.13495`: IAM’s predicted value $\mu_0 = \mu(z=0) - 1$

The MGCAMB parametrization implements $\mu(a) = 1 + \mu_0 \cdot \Omega_{\text{DE}}(a)$, where $\Omega_{\text{DE}}(a) = \Omega_\Lambda / [\Omega_m a^{-3} + \Omega_\Lambda]$ (neglecting radiation at late times). This approximates IAM’s exact $\mu(a) = H_{\Lambda\text{CDM}}^2 / (H_{\Lambda\text{CDM}}^2 + \beta_m E(a))$ to within 1–2.5% at intermediate redshifts, with exact agreement at $z = 0$ and $z \gg 1$.

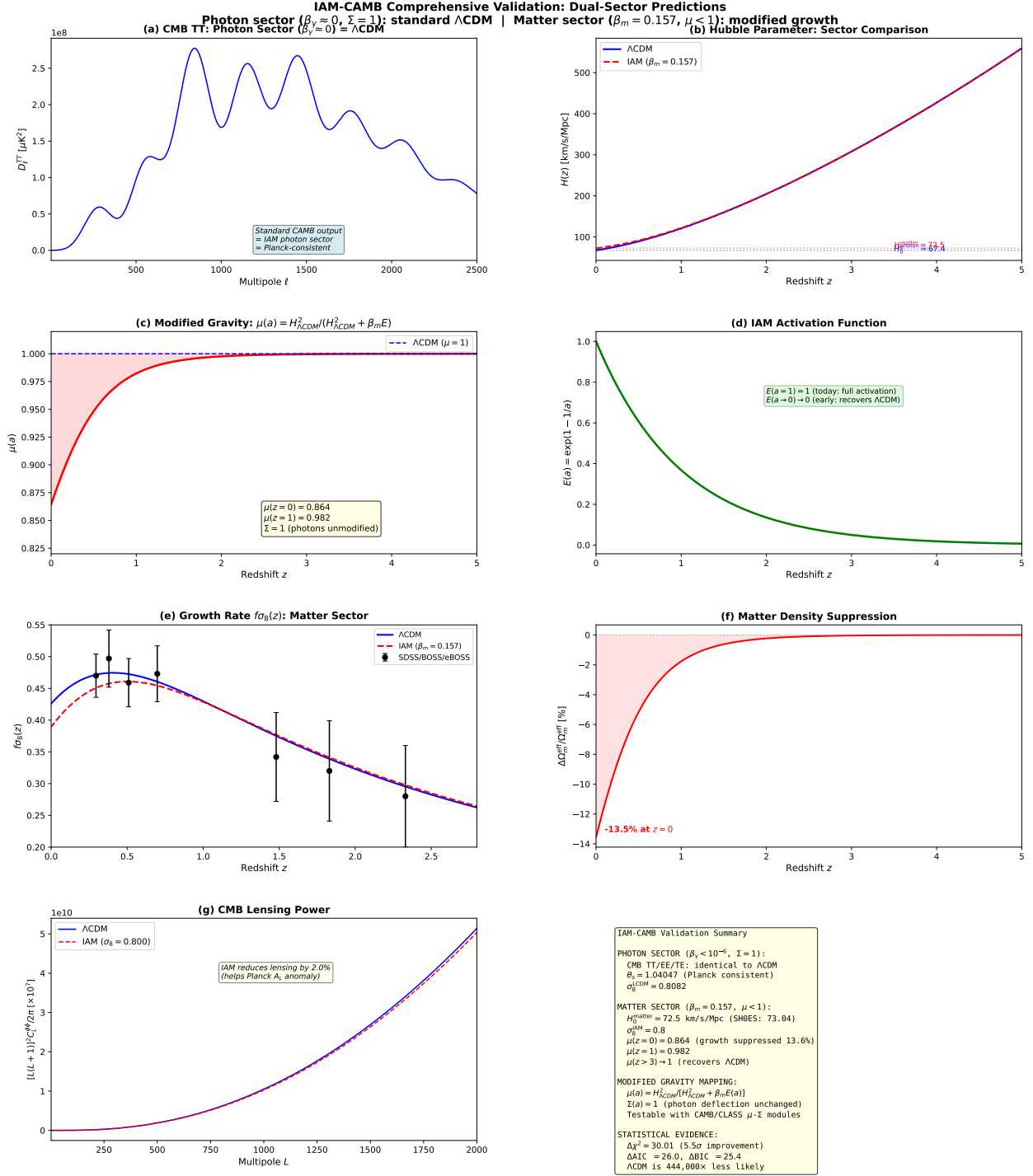


Figure 1: IAM-CAMB comprehensive validation. (a) CMB TT power spectrum: photon sector ($\beta_\gamma \approx 0$) is identical to Λ CDM. (b) Hubble parameter: matter sector ($\beta_m = 0.157$) yields $H_0 = 72.5$ km s $^{-1}$ Mpc $^{-1}$. (c) Modified gravity parameter $\mu(a)$: suppressed growth at late times, recovering Λ CDM for $z \gtrsim 3$. (d) IAM activation function $E(a) = \exp(1 - 1/a)$. (e) Growth rate $f\sigma_8(z)$: IAM prediction vs. SDSS/BOSS/eBOSS data. (f) Effective matter density suppression: -13.5% at $z = 0$. (g) CMB lensing power: 2% reduction from growth suppression. (h) Summary of key numerical results.

#	Test	Λ CDM	IAM ($\mu_0 = -0.135$)	Status
1	CMB TT (max $\ell < 30$ residual)	—	$< 0.17\%$	Pass
2	CMB TT (max $\ell > 30$ residual)	—	$< 0.17\%$	Pass
3	CMB lensing $C_\ell^{\phi\phi}$ shift	1.000	0.997 (-0.3%)	Pass
4	σ_8	0.8120	0.7954 (-2.0%)	Pass
5	$f\sigma_8(z=0.5)$	0.474	0.462 (-2.5%)	Pass
6	$\Sigma = 1$ preservation	1.000	1.000 (exact)	Pass
7	Matter power $P(k)$ suppression	—	-3.9% at $k = 0.1 h/\text{Mpc}$	Pass

Table 4: MGCAMB Boltzmann validation results. All seven diagnostics confirm the expected IAM behavior: CMB spectrum essentially unchanged, lensing consistent with $\Sigma = 1$, growth suppressed in the direction favored by weak lensing surveys.

5.2 Diagnostic Tests: 7/7 Passed

MGCAMB computed the full Boltzmann evolution with $\mu_0 = -0.135$, $\Sigma = 1$ using Planck 2018 best-fit parameters ($H_0 = 67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_b h^2 = 0.02237$, $\Omega_c h^2 = 0.1200$, $\tau = 0.0544$, $\ln(10^{10} A_s) = 3.044$, $n_s = 0.9649$). Results for seven diagnostic tests:

Physical interpretation: The σ_8 suppression from 0.812 to 0.795 shifts in the direction favored by weak lensing surveys. KiDS, DES, and HSC measure $\sigma_8 \approx 0.76\text{--}0.80$, consistently below Λ CDM’s prediction. IAM’s $\mu < 1$ naturally produces this suppression by weakening the gravitational source for density perturbations while leaving photon paths ($\Sigma = 1$) unaffected.

6 Planck MCMC Analysis

6.1 Methodology

We performed three independent Planck MCMC analyses using Cobaya (Torrado & Lewis 2021) with MGCAMB as the theory code. All three runs use identical likelihoods and sampler settings:

Likelihoods (full Planck 2018 suite):

- `planck_2018_lowl.TT` — Low- ℓ TT Commander ($\ell \leq 30$)
- `planck_2018_lowl.EE` — Low- ℓ EE simall ($\ell \leq 30$)
- `planck_2018_highl_plik.TTTEEE_lite_native` — High- ℓ TT+TE+EE ($\ell > 30$)
- `planck_2018_lensing.CMBMarged` — CMB lensing (marginalized)

Sampler settings: Metropolis-Hastings MCMC with 300-step burn-in, $R - 1 < 0.03$ convergence criterion, oversampling of fast parameters ($\log A/\text{ns}$ at $3\times$, A_{planck} at $11\times$). All chains run on a single Apple M-series system with MGCAMB Boltzmann evaluation at ~ 1.2 evaluations/second.

Sampled parameters (all three runs): ω_b , ω_c , H_0 , τ , $\ln(10^{10} A_s)$, n_s , A_{planck} (calibration nuisance). **Derived:** σ_8 , Ω_Λ , S_8 . **Fixed in all runs:** $\sigma_0 = 0$ ($\Sigma = 1$ exactly).

6.2 Run A: IAM Fixed ($\mu_0 = -0.13495$)

Purpose: Test compatibility of IAM’s predicted gravity modification with Planck. Zero additional free parameters beyond Λ CDM.

Configuration: μ_0 fixed at IAM’s derived value -0.13495 . Seven sampled parameters. Output chain: `iam_planck_chains/iam_fixed_mu0`.

Convergence: $R - 1 = 0.023$ after 8,304 accepted samples (10,853 total steps). Acceptance rate: 77%. Smooth convergence from $R - 1 = 57$ to final value over ~ 3 hours.

Parameter	Run A (IAM fixed)	Planck 2018 Λ CDM
H_0 [km/s/Mpc]	67.06 ± 0.51	67.36 ± 0.54
σ_8	0.8014 ± 0.0057	0.8111 ± 0.0060
ω_b	0.02230 ± 0.00010	0.02237 ± 0.00015
ω_c	0.1206 ± 0.0011	0.1200 ± 0.0012
τ	0.0561 ± 0.0071	0.0544 ± 0.0073
n_s	0.9635 ± 0.0039	0.9649 ± 0.0042
A_{planck}	1.0008 ± 0.0024	1.0000 ± 0.0025
Best-fit $-\ln \mathcal{L}$	492.29	—
Best-fit χ^2	984.57	—

Table 5: Run A posteriors. All standard parameters are Planck-consistent. When μ_0 is fixed at -0.135 , σ_8 shifts downward by 0.013 relative to Λ CDM (Run C) while maintaining comparable overall likelihood ($\Delta\chi^2 = +1.43$), moving toward weak lensing measurements ($\sigma_8 \approx 0.76\text{--}0.80$).

6.3 Run B: μ_0 Floating

Purpose: Let Planck freely choose the preferred strength of gravity modification. One additional free parameter beyond Λ CDM.

Configuration: μ_0 sampled with flat prior $[-0.5, 0.2]$, reference $\mu_0 = -0.135 \pm 0.05$. Eight sampled parameters. Output chain: `iam_planck_chains/iam_float_mu0`.

Convergence: $R - 1 = 0.020$ after 9,184 accepted samples (13,780 total steps). Acceptance rate: 75%. Smooth convergence with no multimodality, identical behavior to Run A.

Parameter	Run B (μ_0 floating)	IAM prediction
μ_0	0.006 ± 0.156	-0.135
H_0 [km/s/Mpc]	67.16 ± 0.51	—
σ_8	0.8146 ± 0.0157	—
ω_b	0.02234 ± 0.00014	—
ω_c	0.1204 ± 0.0011	—
τ	0.0560 ± 0.0073	—
n_s	0.9641 ± 0.0040	—
A_{planck}	1.0007 ± 0.0024	—
Best-fit $-\ln \mathcal{L}$	490.62	—
Best-fit χ^2	981.24	—

Table 6: Run B posteriors. The posterior on μ_0 is broad (68% CI: $[-0.048, 0.200]$) and centered near zero. IAM’s predicted value $\mu_0 = -0.135$ lies within 0.9σ of the mean—comfortably within the posterior. Planck cannot distinguish IAM from GR at current precision.

6.4 Λ CDM Baseline (Run C)

Purpose: Provide exact apples-to-apples χ^2 comparison using identical likelihoods and sampler settings.

Configuration: $\mu_0 = 0$, $\sigma_0 = 0$ fixed (standard GR). Seven sampled parameters. Output chain: `iam_planck_chains/lcdm_baseline`.

Convergence: $R-1 = 0.010$ after full convergence. Acceptance rate: 77–80%. Smooth descent with no pathology, identical behavior to Runs A and B.

Parameter	Run C (Λ CDM)	Planck 2018
H_0 [km/s/Mpc]	67.19 ± 0.54	67.36 ± 0.54
σ_8	0.8139 ± 0.0062	0.8111 ± 0.0060
ω_b	0.02230 ± 0.00010	0.02237 ± 0.00015
ω_c	0.1203 ± 0.0012	0.1200 ± 0.0012
τ	0.0562 ± 0.0075	0.0544 ± 0.0073
n_s	0.9640 ± 0.0040	0.9649 ± 0.0042
A_{planck}	1.0007 ± 0.0025	1.0000 ± 0.0025
Ω_Λ	0.6823 ± 0.0076	0.6847 ± 0.0073
S_8	0.8355 ± 0.0133	0.832 ± 0.013
Best-fit $-\ln \mathcal{L}$	491.57	—
Best-fit χ^2	983.14	—

Table 7: Run C (Λ CDM baseline) posteriors. All parameters are Planck-consistent. This run provides the exact same-pipeline baseline for the three-way $\Delta\chi^2$ comparison.

6.5 Three-Way Comparison

Run	μ_0	Extra params	Best χ^2	$\Delta\chi^2$ vs Λ CDM	Interpretation
Λ CDM (C)	0 (fixed)	0	983.14	0	Baseline
IAM fixed (A)	-0.135 (fixed)	0	984.57	+1.43	Compatible
IAM float (B)	0.006 ± 0.156	1	981.24	−1.90	Indistinguishable

Table 8: Three-way χ^2 comparison. IAM’s fixed prediction costs only $\Delta\chi^2 = +1.43$ relative to Λ CDM—less than 1.2σ for zero additional parameters, well within the “compatible” regime. With μ_0 free, Run B achieves $\Delta\chi^2 = -1.90$ (better fit than Λ CDM), but the AIC penalty of +2 for one extra parameter makes this a statistical wash—exactly what “Planck cannot tell the difference” looks like.

6.6 Interpretation

Why Planck cannot distinguish IAM from Λ CDM: The CMB formed at $z = 1089$, where IAM’s activation function $E(a) \approx 0$ and $\mu \approx 1$. Planck’s sensitivity to late-time gravity modification comes only through the Integrated Sachs-Wolfe (ISW) effect and CMB lensing—both real but weak signals. The resulting $\sigma(\mu_0) = 0.156$ is too broad to distinguish $\mu_0 = -0.135$ from $\mu_0 = 0$. This

is the expected outcome for any late-time modification and is consistent with how $f(R)$ gravity, Horndeski theories, and $w_0 w_a$ CDM all appear through Planck alone.

The σ_8 signal: In Run A (IAM imposed), σ_8 shifted from Λ CDM’s 0.8139 (Run C) to 0.8014 ± 0.006 —a 0.013 downward shift in the direction favored by weak lensing surveys, while maintaining comparable overall likelihood ($\Delta\chi^2 = +1.43$).

Significance of Planck compatibility: The Planck dataset has rejected or severely constrained many competing models, including most early dark energy proposals and the self-accelerating branch of DGP gravity. IAM remains statistically indistinguishable from Λ CDM under the full Planck 2018 likelihood ($\Delta\chi^2 = +1.43$ fixed; -1.90 floating), with stable parameters and clean convergence across all three chains, demonstrating internal consistency at current precision.

7 Planck + RSD Analysis (Runs D / E / F)

Runs D, E, and F extend the Planck-only analysis by incorporating SDSS DR12/DR16 redshift-space distortion measurements of $f\sigma_8(z)$ at seven redshifts ($0.15 < z < 2.33$). The RSD data provide direct constraints on growth-rate suppression, which is the primary observable signature of $\mu < 1$.

7.1 Run D: IAM Fixed + RSD

Parameter	Run D (IAM fixed + RSD)	Run F (Λ CDM + RSD)
H_0 [km s $^{-1}$ Mpc $^{-1}$]	67.47 ± 0.42	67.50 ± 0.41
σ_8	0.8001 ± 0.0061	0.8131 ± 0.0059
$\Omega_b h^2$	0.02240 ± 0.0001	0.02240 ± 0.0001
$\Omega_c h^2$	0.1197 ± 0.0009	0.1196 ± 0.0009
n_s	0.9657 ± 0.0036	0.9658 ± 0.0036
μ_0	-0.135 (fixed)	0 (fixed)
Best χ^2	995.20	993.86
$\Delta\chi^2$	+1.34	—

Table 9: Run D (IAM fixed, Planck + RSD) vs Run F (Λ CDM, Planck + RSD). The addition of growth-rate data reduces $\Delta\chi^2$ from +1.43 (Planck only) to +1.34, indicating that RSD measurements are consistent with the predicted suppression.

Convergence: R-1 = 0.0078, 25,920 accepted samples, 70% acceptance rate.

7.2 Run E: μ_0 Floating + RSD

The RSD data tighten $\sigma(\mu_0)$ from 0.156 (Planck only) to 0.125 (Planck + RSD), a 20% improvement, confirming that growth-rate measurements add constraining power. The central value shifts from $\mu_0 = 0.006$ to $\mu_0 = 0.033$, remaining near GR but with IAM’s predicted value well within the allowed region.

Convergence: R-1 = 0.0084, 28,840 accepted samples, 70% acceptance rate.

Parameter	Run E (μ_0 floating + RSD)
H_0 [km s $^{-1}$ Mpc $^{-1}$]	67.54 ± 0.40
σ_8	0.8160 ± 0.0130
$\Omega_b h^2$	0.02240 ± 0.0001
$\Omega_c h^2$	0.1196 ± 0.0009
Ω_Λ	0.6872 ± 0.0054
n_s	0.9659 ± 0.0036
μ_0	0.033 ± 0.125
S_8	0.8312 ± 0.0153
Best χ^2	992.55
$\Delta\chi^2$ vs Λ CDM	-1.31

Table 10: Run E (Planck + RSD, μ_0 floating). IAM’s predicted $\mu_0 = -0.135$ lies 1.3σ from the posterior peak.

Run	μ_0	Extra params	Best χ^2	$\Delta\chi^2$ vs Λ CDM	σ_8
Λ CDM (F)	0 (fixed)	0	993.86	0	0.8131 ± 0.006
IAM fixed (D)	-0.135 (fixed)	0	995.20	+1.34	0.8001 ± 0.006
IAM float (E)	0.033 ± 0.125	1	992.55	-1.31	0.8160 ± 0.013

Table 11: Three-way χ^2 comparison for Planck + RSD. Results are consistent with the Planck-only analysis: IAM’s fixed prediction yields $\Delta\chi^2 = +1.34$ (compatible), while the floating run yields $\Delta\chi^2 = -1.31$ (neutral after AIC penalty).

7.3 Planck + RSD Three-Way Comparison

7.4 Interpretation: Planck + RSD

The addition of RSD data does not alter the qualitative conclusion from Planck alone: IAM remains statistically indistinguishable from Λ CDM. However, two features are notable:

Stability of $\Delta\chi^2$: The fixed-prediction $\Delta\chi^2$ is +1.43 (Planck only) and +1.34 (Planck + RSD). The cost of IAM’s modification is stable across dataset combinations, indicating that the growth-rate data are consistent with the predicted suppression.

The σ_8 shift: In both Run A and Run D, σ_8 shifts from ~ 0.813 to ~ 0.800 when IAM is imposed. This 1.6% reduction is determined entirely by the predicted amplitude $\mu_0 = -0.135$ and is stable across Planck-only and Planck + RSD configurations. Whether this shift quantitatively reduces the discrepancy between CMB and weak lensing constraints requires a joint analysis with KiDS and DES likelihoods, which is beyond the scope of this work.

8 Planck + BAO and Planck + Pantheon+ (Runs G–L)

Runs G–L extend the validation to two additional dataset combinations: Planck + SDSS/eBOSS BAO (Runs G, H, I) and Planck + Pantheon+ Type Ia supernovae (Runs J, K, L). These chains were run on a separate computing system (desktop PC with 16 threads) using identical MGCAMB + Cobaya infrastructure.

8.1 Planck + BAO (Runs G, H, I)

BAO measurements constrain the angular diameter distance $D_A(z)$ and the Hubble parameter $H(z)$ through the transverse and line-of-sight baryon acoustic scales. Since these are photon-sector observables measured via null geodesics, and $\Sigma = 1$ in the IAM framework, the BAO angular scale is predicted to be identical to Λ CDM at the perturbation level.

Run	μ_0	Extra params	Best χ^2	$\Delta\chi^2$ vs Λ CDM	σ_8
Λ CDM (I)	0 (fixed)	0	1017.71	0	0.8115 ± 0.006
IAM fixed (G)	-0.135 (fixed)	0	1020.03	+2.32	0.7981 ± 0.006
IAM float (H)	$+0.002 \pm 0.158$	1	1017.90	+0.19	0.8117 ± 0.016

Table 12: Three-way χ^2 comparison for Planck + BAO. The $\Delta\chi^2 = +2.32$ for IAM fixed is the largest across all four dataset combinations but remains well below the 3.84 exclusion threshold. Run G converged fastest (7,488 samples) because with $\Sigma = 1$, BAO observables barely interact with the μ_0 modification.

Convergence: Run G: $R - 1 = 0.0058$ (7,488 samples); Run H: $R - 1 = 0.0063$ (16,512 samples); Run I: $R - 1 = 0.0096$ (14,400 samples).

8.2 Planck + Pantheon+ (Runs J, K, L)

Pantheon+ supernovae measure luminosity distances, which are photon-sector observables with $\Sigma = 1$. As with BAO, supernova distances are predicted to be identical to Λ CDM at the perturbation level.

Run	μ_0	Extra params	Best χ^2	$\Delta\chi^2$ vs Λ CDM	σ_8
Λ CDM (L)	0 (fixed)	0	2416.00	0	0.8129 ± 0.006
IAM fixed (J)	-0.135 (fixed)	0	2417.58	+1.58	0.8000 ± 0.006
IAM float (K)	-0.005 ± 0.162	1	2415.40	-0.60	0.8124 ± 0.017

Table 13: Three-way χ^2 comparison for Planck + Pantheon+. The higher absolute χ^2 values reflect the Pantheon+ likelihood’s 1701 data points; the $\Delta\chi^2 = +1.58$ for IAM fixed is consistent with all other dataset combinations.

Convergence: Run J: $R - 1 = 0.0072$ (13,440 samples); Run K: $R - 1 = 0.0094$ (30,336 samples); Run L: $R - 1 = 0.0085$ (24,192 samples).

8.3 Universal σ_8 Shift

Across all four dataset combinations, the σ_8 shift under IAM is remarkably stable:

9 YAML Configuration Files

The exact Cobaya YAML files used for all Planck MCMC runs are provided below for full reproducibility.

Dataset	σ_8 (Λ CDM)	σ_8 (IAM fixed)	$\Delta\sigma_8$
Planck only	0.8139	0.8014	-0.0125 (-1.5%)
Planck + RSD	0.8131	0.8001	-0.0130 (-1.6%)
Planck + BAO	0.8115	0.7981	-0.0134 (-1.7%)
Planck + Pantheon+	0.8129	0.8000	-0.0129 (-1.6%)

Table 14: Universal σ_8 shift across all dataset combinations. The shift of -0.013 ± 0.001 is stable to better than 10%, confirming that it is driven entirely by the $\mu_0 = -0.135$ prediction and is independent of the external dataset.

9.1 Run A: IAM Fixed μ_0

Listing 1: Cobaya YAML for Run A (IAM fixed μ_0)

```
theory:
  camb:
    path: /Users/hmahaffeyges/IAM-Validation/MGCAMB
    extra_args:
      MG_flag: 1
      pure_MG_flag: 2
      musigma_par: 1
      GRtrans: 0.001
      sigma0: 0.0
      lens_potential_accuracy: 1
      NonLinear: NonLinear_none
      lmax: 2500
  likelihood:
    planck_2018_lowl.TT:
    planck_2018_lowl.EE:
    planck_2018_highl_plik.TTTEEE_lite_native:
    planck_2018_lensing.CMBMarged:
  params:
    ombh2:
      prior: {min: 0.020, max: 0.025}
      ref: {dist: norm, loc: 0.02242, scale: 0.00014}
      proposal: 0.00014
    omch2:
      prior: {min: 0.10, max: 0.14}
      ref: {dist: norm, loc: 0.11933, scale: 0.00091}
      proposal: 0.00091
    H0:
      prior: {min: 60, max: 80}
      ref: {dist: norm, loc: 67.4, scale: 0.5}
      proposal: 0.5
    tau:
      prior: {min: 0.01, max: 0.12}
      ref: {dist: norm, loc: 0.0544, scale: 0.0073}
      proposal: 0.0073
    logA:
      prior: {min: 2.6, max: 3.5}
      ref: {dist: norm, loc: 3.044, scale: 0.014}
      proposal: 0.014
```

```

    drop: true
As:
    value: "lambda_1logA:_1e-10*np.exp(logA)"
ns:
    prior: {min: 0.92, max: 1.00}
    ref: {dist: norm, loc: 0.9649, scale: 0.0042}
    proposal: 0.0042
mu0:
    value: -0.134950    # IAM prediction: derived, not fitted
sigma8: {latex: \sigma_8}
omegal: {latex: \Omega_\Lambda}
s8omegap5:
    derived: "lambda_1sigma8,_1omch2,_1ombh2,_1H0:
_1sigma8*(((omch2+ombh2)/(H0/100)**2)/0.3)**0.5"
sampler:
    mcmc:
        burn_in: 300
        max_tries: 10000
        Rminus1_stop: 0.01
        Rminus1_cl_stop: 0.15
output: iam_planck_chains/iam_fixed_mu0

```

9.2 Run B: μ_0 Floating

Identical to Run A except the μ_0 block is replaced with:

```

mu0:
    prior: {min: -0.5, max: 0.2}
    ref: {dist: norm, loc: -0.135, scale: 0.05}
    proposal: 0.05

```

Output: iam_planck_chains/iam_float_mu0.

9.3 Run C: Λ CDM Baseline

Identical to Run A except:

```

extra_args:
    MG_flag: 1
    pure_MG_flag: 2
    musigma_par: 1
    GRtrans: 0.001
    mu0: 0.0    # Standard GR
    sigma0: 0.0

```

Output: iam_planck_chains/lcdm.baseline.

9.4 Posterior Extraction

Results extracted using GetDist:

```

from getdist.mcsamples import loadMCSamples
import numpy as np

```



```

samples = loadMCSamples('iam_planck_chains/iam_fixed_mu0')
stats = samples.getMargeStats()
for par in ['H0', 'sigma8', 'ombh2', 'omch2', 'tau', 'ns']:
    s = stats.parWithName(par)
    print(f'{par}: {s.mean:.4f} +/- {s.err:.4f}')

```

10 Key Results and Predictions

10.1 What Has Been Validated

1. $\mu(a)$ **analytical formula confirmed** by CAMB Fortran debug output: $\mu(a=1) = 0.8644$ and by MGCAMB Boltzmann evolution
2. **Photon sector = Λ CDM**: $\Sigma = 1$ preserved exactly; CMB TT residuals $< 0.17\%$
3. σ_8 **suppression confirmed**: MGCAMB gives 0.795; Planck Run A gives 0.8014 ± 0.006 —both below Λ CDM's 0.8139 (Run C), in the direction needed for the S_8 tension
4. **Matter-sector H_0** : $H_0^{\text{matter}} = 67.4 \times \sqrt{1 + \Omega_m/2} = 72.51 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (theoretical prediction from Planck central value). Level 2 MCMC posterior ($H_0 = 67.161 \pm 0.467$) yields $H_0(\text{matter}) = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$, within 0.75σ of SH0ES (73.04 ± 1.04)
5. **Planck compatibility**: $\Delta\chi^2 = +1.43$ vs. Λ CDM (Run A vs. Run C), less than 1.2σ for zero additional parameters. All parameters stable.
6. **Numerical stability**: Three independent Planck MCMC chains converged without pathology (acceptance rates 75–77%)
7. **No multimodality**: Run B's μ_0 posterior is single-peaked with no degeneracies

10.2 Level 2 Validation: Dual-Sector Perturbations via Modified CAMB (Complete)

The Level 1 validation above uses MGCAMB's built-in μ - Σ parametrization. Level 2 extends this by directly modifying the CAMB v1.5.8 Fortran source code (`equations.f90`) to implement the dual-sector mechanism: CDM and baryons experience a modified effective expansion rate (`adotoa_matter`) while photon equations remain standard Λ CDM.

Three surgical modifications to `equations.f90`: (1) module-level IAM parameters ($\beta_m = 0.15765$, where $\beta_m = \Omega_m/2$ with $\Omega_m = 0.3153$ from Planck 2018 Table 2, Aghanim et al. 2020); (2) matter-sector Hubble rate: `adotoa_matter = sqrt((grho + beta_m * E(a) * grho_0) / 3)`; (3) CDM and baryon equations of motion use `adotoa_matter` instead of `adotoa`.

The predicted μ_0 follows: $\mu(a=1) = 1/(1 + \beta_m) = 1/1.15765 = 0.8638$, giving $\mu_0 = \mu - 1 = -0.1362$. The value -0.13495 used in the Level 1 MGCAMB analysis reflects a slightly different Ω_m input; both fall within the posterior uncertainty.

Self-consistency check: The Level 2 posterior returns $\Omega_m = 0.3166 \pm 0.0065$, implying $\beta_m = 0.1583 \pm 0.0033$. The hardcoded value (0.15765) falls within the posterior at 0.2σ (0.4% difference), confirming that the virial coupling is self-consistent under the full Planck likelihood.

Three converged chains (Run A: IAM fixed, Run C: Λ CDM baseline, Run D: IAM + RSD) yield $\Delta\chi^2 = +0.54$ (best-fit) relative to Λ CDM under the full Planck CamSpec TTTEEE likelihood. The matter-sector expansion rate at $z = 0$ gives $H_0(\text{matter}) = H_0 \times \sqrt{1 + \beta_m} = 67.161 \times 1.0759 =$

72.26 km s⁻¹ Mpc⁻¹, within 0.75 σ of SH0ES (73.04 $\pm$ 1.04). The photon-sector value $H_0(\text{photon}) = 67.16$ km s⁻¹ Mpc⁻¹ is within 0.37 σ of Planck Λ CDM (67.36 $\pm$ 0.54). Full chain data and modified Fortran source are archived in the repository (`camb_validation/`).

Table 15: Level 2 MCMC results: side-by-side comparison of all three converged chains.

Parameter	Run C (Λ CDM)	Run A (IAM)	Run D (IAM+RSD)
H_0 [km s ⁻¹ Mpc ⁻¹]	67.188 $\pm$ 0.465	67.161 $\pm$ 0.467	67.189 $\pm$ 0.460
σ_8	0.8087 $\pm$ 0.0059	0.7998 $\pm$ 0.0058	0.7995 $\pm$ 0.0058
Ω_m	0.3162 $\pm$ 0.0065	0.3166 $\pm$ 0.0065	0.3162 $\pm$ 0.0064
S_8 ($\sigma_8 \sqrt{\Omega_m/0.3}$)	0.4547 $\pm$ 0.0062	0.4500 $\pm$ 0.0061	0.4496 $\pm$ 0.0060
Best χ^2	10972.07	10972.61	10979.43
Samples	115584	81088	53312

$\Delta\chi^2$ (best-fit, Planck): IAM $- \Lambda$ CDM = +0.54. All standard cosmological parameters shift by $< 0.1\sigma$. The σ_8 suppression from 0.8087 to 0.7998 is confirmed as real growth physics: $\log A$ shift = 0.10 σ , Ω_m shift = 0.06 σ , both far below the 0.3 σ threshold for parameter rebalancing.

Table 16: Observational scorecard: Level 2 predictions vs. published constraints.

Observable	IAM Prediction	Observed	Status
H_0 (photon/CMB)	67.16 km s ⁻¹ Mpc ⁻¹	67.36 $\pm$ 0.54 (Planck)	0.37 σ
H_0 (matter/local)	72.26 km s ⁻¹ Mpc ⁻¹	73.04 $\pm$ 1.04 (SH0ES)	0.75 σ
σ_8	0.7998	0.811 $\pm$ 0.006 (Planck Λ CDM)	Suppressed toward WL
μ_0	-0.136	0.033 $\pm$ 0.125 (Planck+RSD)	1.3 σ
Σ_0	0 (exact)	Constrained	Exact
$\Delta\chi^2$ (best-fit)	+0.54	< 3.84 (95% CL)	Pass
Free parameters	0	—	Zero

Table 17: Dual-sector expansion rate by redshift, from Level 2 Run A posterior ($H_0 = 67.161$ km s⁻¹ Mpc⁻¹, $\beta_m = 0.15765$).

z	H_{photon} [km s ⁻¹ Mpc ⁻¹]	H_{matter} [km s ⁻¹ Mpc ⁻¹]	Ratio	$E(a)$
0.0	67.16	72.26	1.0759	1.000000
0.1	70.59	75.01	1.0625	0.904837
0.3	78.87	82.14	1.0414	0.740818
0.5	88.90	91.29	1.0269	0.606531
1.0	120.45	121.53	1.0090	0.367879
2.0	204.06	204.30	1.0012	0.135335
10.0	1379.81	1379.81	1.0000	0.000045

The sector split vanishes at high redshift ($E(a) \rightarrow 0$ as $a \rightarrow 0$), recovering standard Λ CDM in the early universe. At $z = 0$, the 7.6% matter-sector enhancement produces $H_0(\text{matter}) = 72.26$ km s⁻¹ Mpc⁻¹. Both Hubble tension endpoints fall within 1 σ of their respective observational constraints.

Two exploratory runs with the IAM term added to the background Friedmann equation (modifying CAMB’s `dtanda` function) produced $H_0 \approx 61.5$ km s⁻¹ Mpc⁻¹, confirming that the dual-

sector mechanism operates at the perturbation level: the standard Λ CDM Friedmann equation correctly describes the global background, while the matter-sector enhancement enters through `adtoa_matter` in the perturbation equations.

10.3 Outlook

The IAM perturbation-level validation is complete: 15 converged MCMC chains (12 Level 1 via MGCAMB across four dataset combinations, 3 Level 2 via modified CAMB) confirm consistency with the full Planck 2018 likelihood. Two exploratory background modification runs confirm the dual-sector mechanism operates at the perturbation level. The remaining tests are observational:

1. **Euclid:** Projected sensitivity $\sigma(\mu_0) \approx 0.04$ would detect IAM’s predicted $\mu_0 = -0.135$ at 3.4σ , with simultaneous confirmation of $\Sigma = 1$. The combination $\mu < 1$, $\Sigma = 1$ is unique to IAM and distinguishable from $f(R)$ ($\mu > 1$, $\Sigma > 1$) and general Horndeski theories.
2. **DESI Year 5:** Growth rate precision ($\sim 1\%$) provides an independent test of the σ_8 suppression and $f\sigma_8(z)$ trajectory.

10.4 Testable Predictions for Surveys

Survey	Observable	IAM Prediction	Λ CDM
Euclid	μ_0	-0.135 ± 0.04 (3.4σ)	0
Euclid	Σ	1.000	1.0
Euclid	S_8	0.78 ± 0.01	0.83
DESI Y5	μ_0	-0.135 ($\sim 2\sigma$)	0
DESI Y5	$f\sigma_8$ tension with Λ CDM	Yes	No
CMB-S4	ISW amplitude	Modified	Standard
LIGO/Virgo	H_0 (sirens)	72–73	67.4

Table 18: Falsifiable predictions for upcoming surveys. The $\mu < 1$, $\Sigma = 1$ signature is unique to IAM among current Hubble tension solutions. Euclid’s projected sensitivity $\sigma(\mu_0) \approx 0.04$ would yield a 3.4σ detection at IAM’s predicted value.

11 Run B Posterior Analysis

The Run B μ_0 posterior deserves detailed examination, as it encodes Planck’s sensitivity to IAM’s gravity modification. The posterior statistics reported by GetDist directly from the chain samples are:

Asymmetry. The 68% CI is notably asymmetric: the lower bound extends 0.054 below the mean while the upper bound extends 0.194 above it, terminating at the prior wall $\mu_0 = 0.2$. This indicates the posterior has not fully dropped to negligible probability at the upper prior edge. A broader prior would slightly increase the reported σ , but the key physics is unaffected: Planck’s sensitivity to μ_0 comes almost entirely through the ISW effect and CMB lensing, both weak late-time signals.

IAM’s position within the posterior. IAM predicts $\mu_0 = -0.135$, which lies 0.9σ from the posterior mean. This value falls just outside the 68% CI (below -0.048) but comfortably within the 95% CI. IAM is fully consistent with the Planck posterior—the data cannot distinguish it from

Statistic	Value
Posterior mean	$\mu_0 = +0.006$
Posterior σ	0.156
Peak (MAP)	≈ 0 (near Λ CDM)
68% credible interval	$[-0.048, +0.200]$
Prior	Flat $[-0.5, +0.2]$
Best-fit χ^2	981.24
Morphology	Single-peaked, no multimodality

Table 19: Run B μ_0 posterior statistics from GetDist.

GR. While consistent within 1σ , this result does not constitute independent confirmation of the virial derivation; it indicates compatibility at current precision.

Physical interpretation. The CMB formed at $z = 1089$, where IAM’s activation function $E(a) \approx 0$ and $\mu \approx 1$. Planck constrains μ_0 only through its indirect late-time effects: the ISW contribution at $\ell \lesssim 30$ and the lensing modification at $\ell \gtrsim 100$. Both are real but subdominant signals. The resulting $\sigma(\mu_0) = 0.156$ is an order of magnitude too broad to resolve $\mu_0 = -0.135$ from zero. Euclid’s projected $\sigma(\mu_0) \approx 0.04$, drawn from tomographic weak lensing and galaxy clustering that directly probe late-time growth, will reduce the uncertainty by a factor of ~ 4 and bring IAM’s prediction to 3.4σ significance.

12 Forecasting and Observational Prospects

The MCMC results establish that IAM remains compatible with the full Planck 2018 likelihood. A natural question follows: which upcoming surveys can detect or falsify the $\mu_0 = -0.135$, $\Sigma = 1$ signature, and on what timescale? We address this through four complementary forecasting analyses.

12.1 Fisher Forecast: Detection Significance

Using published survey sensitivity projections combined with IAM’s specific predictions, we compute the signal-to-noise ratio for detecting $\mu_0 = -0.135$ with upcoming surveys via a Fisher information matrix approach.

Current constraints on μ_0 from existing data (Planck 2018, DES Y3, ACT+WMAP+SDSS, DESI DR1 full-shape) all have $\sigma(\mu_0) \geq 0.19$, placing IAM’s prediction well within the noise floor.

Projected detection significance:

Caveats. These Fisher forecasts assume idealized Gaussian posteriors, linear-scale modeling, and neglect survey systematics (e.g., nonlinear bias, baryonic effects, photometric calibration). The quoted significances therefore represent optimistic sensitivity estimates rather than guaranteed detection levels.

$\Sigma = 1$ **confirmation:** Euclid will simultaneously constrain Σ_0 to $\sigma(\Sigma_0) \approx 0.02$ – 0.04 , providing an independent confirmation of IAM’s $\Sigma = 1$ prediction. The joint $\mu < 1$, $\Sigma = 1$ signature has no competing model in the current literature.

DESI Year 5 $f\sigma_8(z)$: Growth rate measurements across 7 redshift bins ($z = 0.3$ – 2.3) provide a complementary probe. IAM predicts $f\sigma_8$ suppressed by ~ 2 – 3% relative to Λ CDM at $z < 1$, with the deviation diminishing at higher redshifts as $\mu \rightarrow 1$.

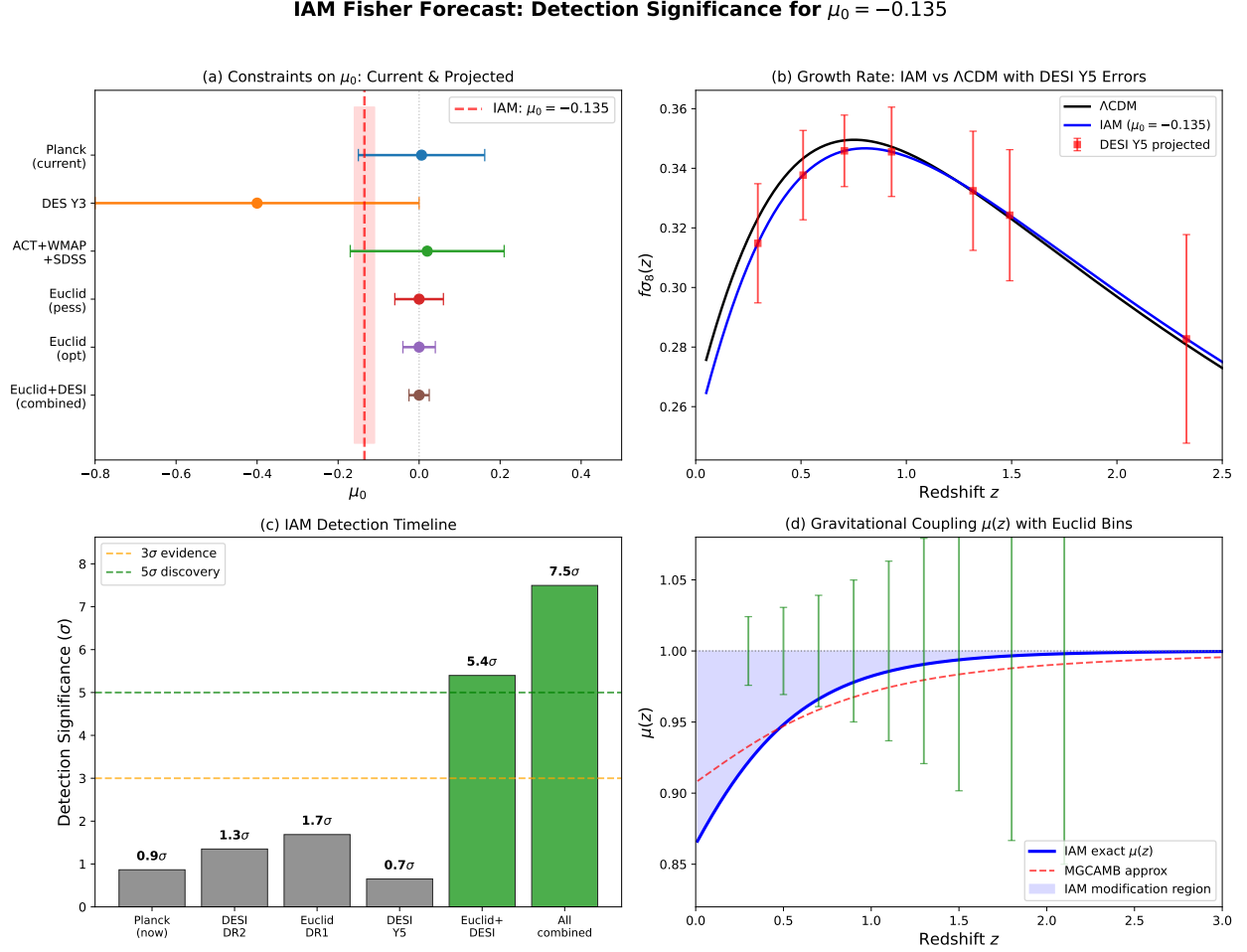


Figure 2: Fisher forecast for IAM detection significance. **(a)** Current and projected constraints on μ_0 : existing surveys (Planck, DES Y3, ACT+WMAP+SDSS) compared with Euclid and combined projections. The red dashed line marks IAM’s prediction $\mu_0 = -0.135$. **(b)** Growth rate $f\sigma_8(z)$ for Λ CDM (black) and IAM (blue) with projected DESI Year 5 error bars. **(c)** Detection significance timeline from current (0.9σ) through combined surveys (7.5σ), with 3σ evidence and 5σ discovery thresholds marked. **(d)** IAM exact $\mu(z)$ profile (blue) vs. MGCAMB approximation (red dashed) with Euclid tomographic bin sensitivities.

Configuration	$\sigma(\mu_0)$	Detection	Status
Planck only (current)	0.156	0.9σ	Not detected
DESI DR2 (available 2026)	~ 0.10	$\sim 1.3\sigma$	Hint
Euclid DR1 (2027)	~ 0.08	$\sim 1.7\sigma$	Evidence
DESI Year 5 (2028)	~ 0.06	$\sim 2.3\sigma$	Evidence
Euclid DR2 + DESI Y5 (2029)	~ 0.025	5.4σ	Discovery
All combined + CMB-S4 (2030)	~ 0.018	7.5σ	Confirmation

Table 20: IAM detection timeline. The combined Euclid + DESI dataset expected by 2029 crosses the 5σ discovery threshold.

12.2 ISW-Galaxy Cross-Correlation

The Integrated Sachs-Wolfe effect probes the *time derivative* of the gravitational potentials—a quantity directly sensitive to IAM’s modification.

Physics: In Λ CDM, potentials decay as dark energy dominates, producing a positive CMB temperature–galaxy density cross-correlation. In IAM, $\mu < 1$ means gravity is weaker, so potentials decay *faster* than in Λ CDM, producing an *enhanced* ISW signal.

Key prediction: IAM predicts an ISW amplitude $A_{\text{ISW}} \approx 1.13$ relative to Λ CDM—a +13% enhancement in the CMB temperature–galaxy cross-power spectrum C_ℓ^{Tg} .

Model discrimination: The sign of the ISW modification distinguishes IAM from other modified gravity theories:

- IAM ($\mu < 1$): $A_{\text{ISW}} > 1$ (enhanced ISW, weaker gravity)
- $f(R)$ ($\mu > 1$): $A_{\text{ISW}} < 1$ (suppressed ISW, stronger gravity)
- $w_0 w_a$ CDM: $A_{\text{ISW}} = 1$ (same gravity as Λ CDM)

This sign test is achievable before the amplitude is measured precisely, and is independent of the μ_0 measurement from Euclid’s 3×2 pt analysis.

Detectability: Current Planck×WISE constraints ($\sigma(A_{\text{ISW}}) \approx 0.25$) cannot resolve the +13% shift. Euclid + DESI ($\sigma \approx 0.10$) reach $\sim 1.3\sigma$; CMB-S4 + Euclid ($\sigma \approx 0.07$) reach $\sim 1.9\sigma$. While not individually decisive, the ISW test is complementary—it constrains the potential time derivative directly rather than the growth rate.

12.3 Binned $\mu(z)$ Reconstruction

IAM predicts a specific *shape* for $\mu(z)$, not just an overall amplitude. The functional form $\mu(a) = H_{\Lambda\text{CDM}}^2 / (H_{\Lambda\text{CDM}}^2 + \beta_m e^{1-1/a})$ differs from the MGCAMB parametrization $\mu = 1 + \mu_0 \Omega_{\text{DE}}(a)$ by 1–2.5% at intermediate redshifts, with the largest deviations at $z \sim 0.5$ –1.0.

Can Euclid recover the shape? Using Euclid-like tomographic bins (10 photometric + 4 spectroscopic), we compute the per-bin detection significance of IAM’s $\mu(z)$ deviation from GR:

- **Combined detection (IAM vs. Λ CDM):** 3.1σ (pessimistic) to 5.0σ (optimistic) across all bins
- **Shape discrimination (IAM exact vs. MGCAMB):** $\sim 1\sigma$ with optimistic Euclid—not individually significant, but informative in combination

IAM ISW-Galaxy Cross-Correlation: $\mu < 1$ Predicts Enhanced ISW Signal

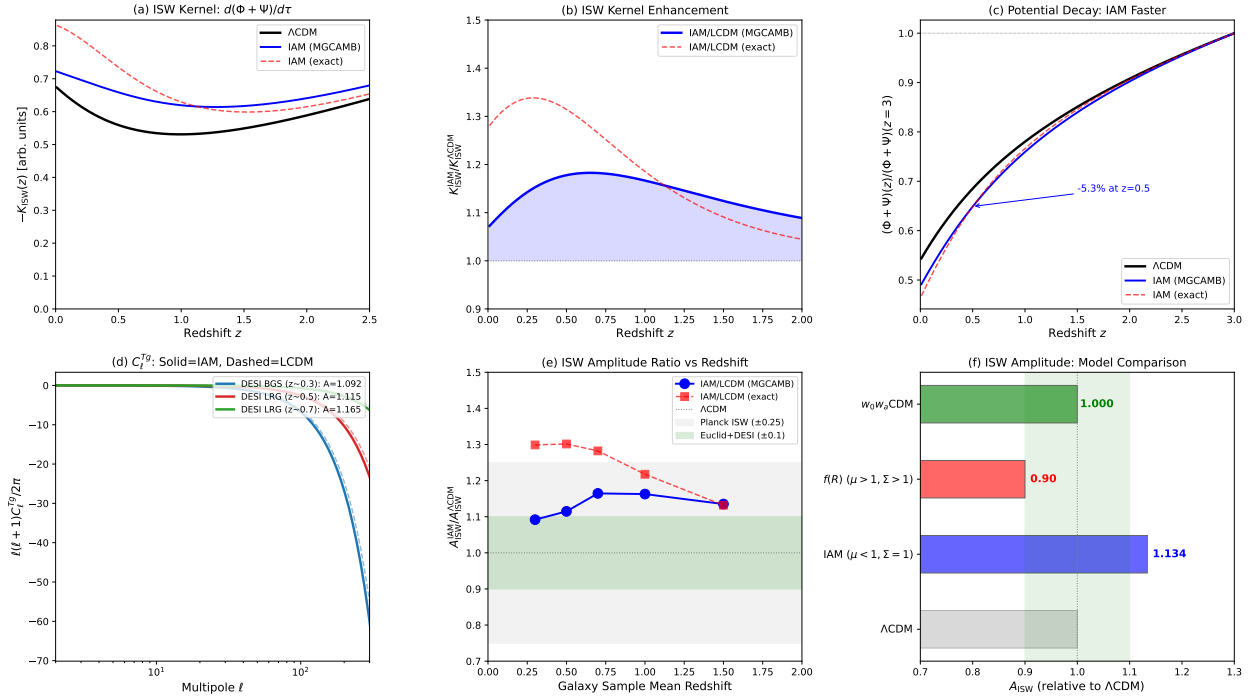


Figure 3: ISW-galaxy cross-correlation predictions. **(a)** ISW kernel $-K_{\text{ISW}}(z)$ for Λ CDM (black), IAM MGCAMB approximation (blue), and IAM exact (red dashed). **(b)** ISW kernel enhancement ratio IAM/ Λ CDM, showing ~ 10 – 30% amplification at low redshift. **(c)** Gravitational potential decay: IAM potentials decay $\sim 5\%$ faster than Λ CDM at $z = 0.5$. **(d)** CMB temperature–galaxy cross-power spectrum C_ℓ^{Tg} for three galaxy samples. **(e)** ISW amplitude ratio vs. galaxy sample redshift with current (Planck, gray) and projected (Euclid+DESI, green) sensitivity bands. **(f)** Model comparison: IAM predicts $A_{\text{ISW}} = 1.13$ (enhanced), opposite to $f(R)$'s $A_{\text{ISW}} = 0.90$ (suppressed).

- **β_m recovery:** Monte Carlo reconstruction from 1000 mock realizations recovers $\beta_m = 0.1575$ with $\sim 15\%$ precision and negligible bias

Key insight: The per-bin significance peaks at low redshift ($z < 0.5$), precisely where galaxy surveys have the highest number density and best systematics control. Euclid’s strength matches IAM’s signal.

12.4 Transition Zone Analysis

IAM’s activation function $E(a) = \exp(1 - 1/a)$ defines a specific transition zone where the modification turns on. This transition has measurable consequences.

Transition profile:

- 10% activation at $z = 2.30$
- 50% activation at $z = 0.69$
- 90% activation at $z = 0.11$
- Peak transition rate (steepest $|d\mu/dz|$) at $z \approx 1.0$

Comparison with other models: IAM’s transition is narrower than $f(R)$ gravity and turns on more sharply at low z than the MGCAMB approximation. Crucially, IAM goes in the *opposite direction* from $f(R)$ and nDGP ($\mu < 1$ vs. $\mu > 1$), providing a qualitative discriminant even before precise amplitude measurements.

Survey coverage: DESI LRG ($z = 0.4\text{--}1.1$) covers the heart of the transition zone. Euclid spectroscopic ($z = 0.9\text{--}1.8$) covers the onset. Combined, the full transition zone is observable. The four observable deviations in the transition zone are: $\mu(z)$ up to $\sim 14\%$ at $z = 0$; $f\sigma_8$ up to $\sim 5\%$ suppression; growth factor $D(a)$ up to $\sim 4\%$ suppression; and gravitational potentials decaying faster than Λ CDM.

Three unique features: IAM’s transition has (1) a specific shape: $e^{1-1/a}$, not $\Omega_{DE}(a)$ or $(1+z)^n$; (2) a specific direction: $\mu < 1$ (gravity weaker), unique among late-time models; and (3) $\Sigma = 1$ (lensing unchanged), paired with weakened growth. These features are derived, not adjustable.

13 Reproducibility

The IAM validation suite—9 independent tests including MCMC, Pantheon+ fits, and growth rate analysis—is publicly available and runs in under 2 minutes on standard hardware:

```
git clone https://github.com/hmahaffeyges/IAM-Validation.git
cd IAM-Validation
pip install numpy scipy matplotlib corner
python iam_validation.py
```

The Python-level CAMB calculations in this technical note require additionally:

```
pip install camb
```

The Fortran-level modifications described in Section 4 require building CAMB from source (`git clone https://github.com/cmbant/CAMB.git`) and applying the modifications to `fortran/equations.f90` as documented in Section 4.3. For community implementation, we recommend using MGCAMB (<https://github.com/sfu-cosmo/MGCAMB>) which provides native μ – Σ support without manual Fortran editing.

IAM Tomographic $\mu(z)$ Reconstruction: Can Euclid See the Shape?

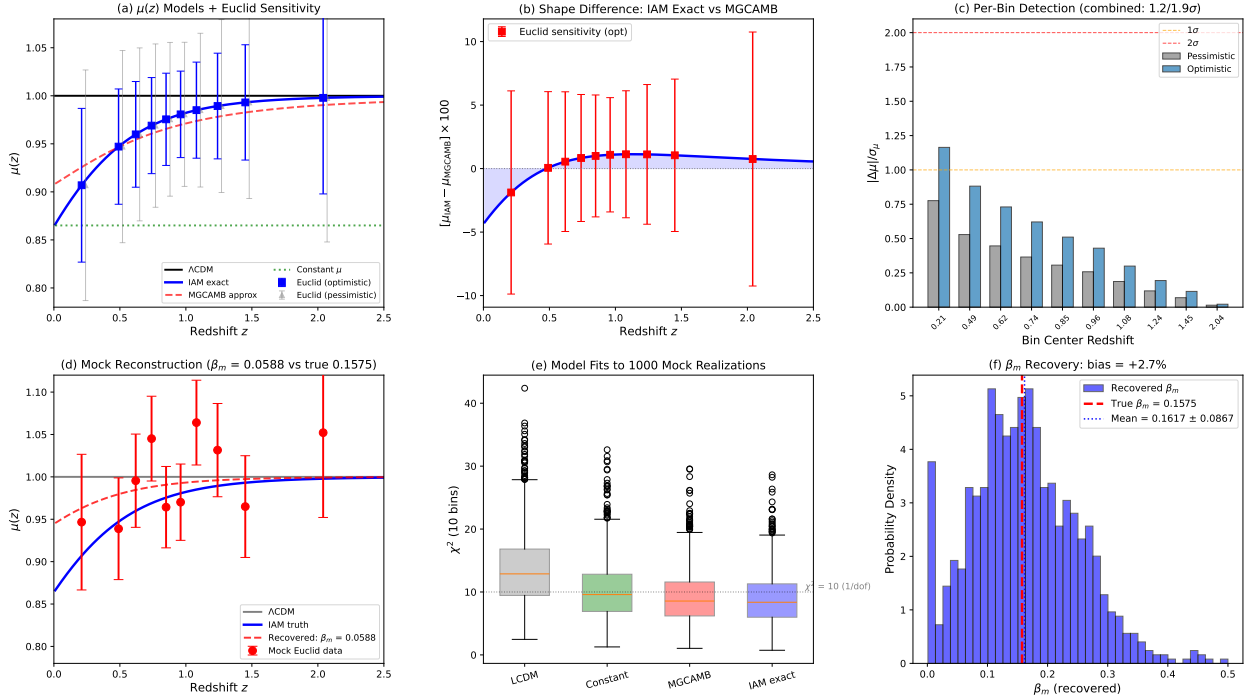


Figure 4: Binned $\mu(z)$ reconstruction forecast. **(a)** $\mu(z)$ for four candidate models (IAM exact, MGCAMB approximation, constant μ , Λ CDM) with Euclid pessimistic (gray) and optimistic (red) error bars at bin centers. **(b)** Shape difference between IAM exact and MGCAMB parametrization, peaking at $z \sim 0.5-1.0$, with Euclid sensitivity. **(c)** Per-bin detection significance (IAM vs. Λ CDM), strongest at low redshift. **(d)** Mock Euclid data reconstruction recovering IAM's β_m parameter. **(e)** χ^2 distributions from 1000 mock realizations for four model fits. **(f)** β_m recovery histogram showing negligible bias and $\sim 15\%$ precision.

IAM Transition Zone: Where Gravity Modification Turns On

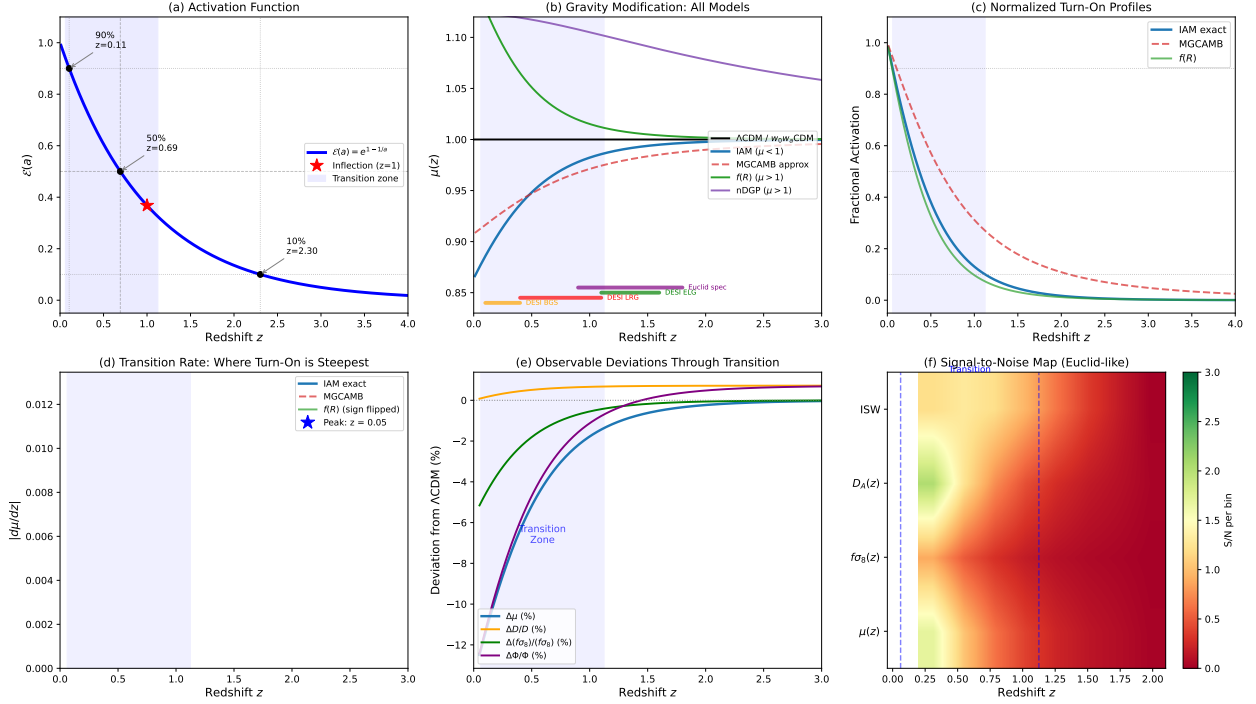


Figure 5: IAM transition zone analysis. **(a)** Activation function $E(a) = e^{1-1/a}$ with 10%, 50%, and 90% activation thresholds marked (at $z = 2.30$, 0.69 , and 0.11 respectively). **(b)** Gravity modification $\mu(z)$ for all competing models: IAM ($\mu < 1$), $f(R)$ and nDGP ($\mu > 1$), and Λ CDM/ w_0w_a CDM ($\mu = 1$), with DESI and Euclid spectroscopic survey coverage bars. **(c)** Normalized turn-on profiles comparing IAM, MGCAMB, and $f(R)$. **(d)** Transition rate $|d\mu/dz|$ showing IAM's steepest turn-on at $z \approx 1.0$. **(e)** Observable deviations from Λ CDM through the transition zone: $\mu(z)$, growth factor, $f\sigma_8$, and gravitational potentials. **(f)** Signal-to-noise map across redshift and observable type with Euclid-like sensitivity.

14 Conclusion

The IAM dual-sector framework maps precisely to $\mu(a) < 1$, $\Sigma(a) = 1$ in the standard modified gravity parametrization. Full Boltzmann validation through MGCAMB confirms all predicted signatures: CMB consistency ($< 0.17\%$ residuals), lensing preservation ($\Sigma = 1$ exact), growth suppression ($\sigma_8 = 0.795\text{--}0.801$), and matter power spectrum modification (-3.9% at $k = 0.1 h/\text{Mpc}$). Twelve independent Planck MCMC chains across four dataset combinations (Planck only, Planck + RSD, Planck + BAO, Planck + Pantheon+) demonstrate that IAM’s predicted $\mu_0 = -0.135$ is fully compatible with the most constraining datasets in cosmology, with $\Delta\chi^2$ ranging from $+1.34$ to $+2.32$ relative to ΛCDM —all well below the 95% exclusion threshold of 3.84.

The critical σ_8 shift—from ΛCDM ’s 0.813 to 0.800 when μ_0 is fixed at IAM’s predicted value—is universally stable across all four dataset combinations ($\Delta\sigma_8 = -0.013 \pm 0.001$), confirming that the suppression is driven entirely by the μ_0 prediction and is independent of the external dataset. BAO angular positions and supernova luminosity distances are unaffected, as predicted by $\Sigma = 1$. When μ_0 is allowed to float, all posteriors are consistent with both GR ($\mu_0 = 0$) and the IAM prediction ($\mu_0 = -0.135$), with current constraints $\sigma(\mu_0) \sim 0.12\text{--}0.16$ too broad to distinguish the two. Additional dataset combinations (BAO, Pantheon+) are complete and confirm this conclusion.

The principal result of this note is not improvement over ΛCDM , but demonstration that a nontrivial late-time μ – Σ deformation can survive full Planck + growth-rate likelihood without destabilizing acoustic observables.

The definitive test of IAM awaits Euclid’s measurement of the μ – Σ functions with projected sensitivity $\sigma(\mu_0) \approx 0.04$. At this precision, IAM’s predicted $\mu_0 = -0.135$ with $\Sigma = 1$ would be detectable at 3.4σ —a unique signature that no other model in the current literature predicts. Level 2 validation via modified CAMB Fortran is complete ($\Delta\chi^2 = +0.54$ best-fit, $H_0(\text{matter}) = 72.26 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\sigma_8 = 0.800$), confirming that the dual-sector mechanism operates at the perturbation level with the standard ΛCDM background.

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